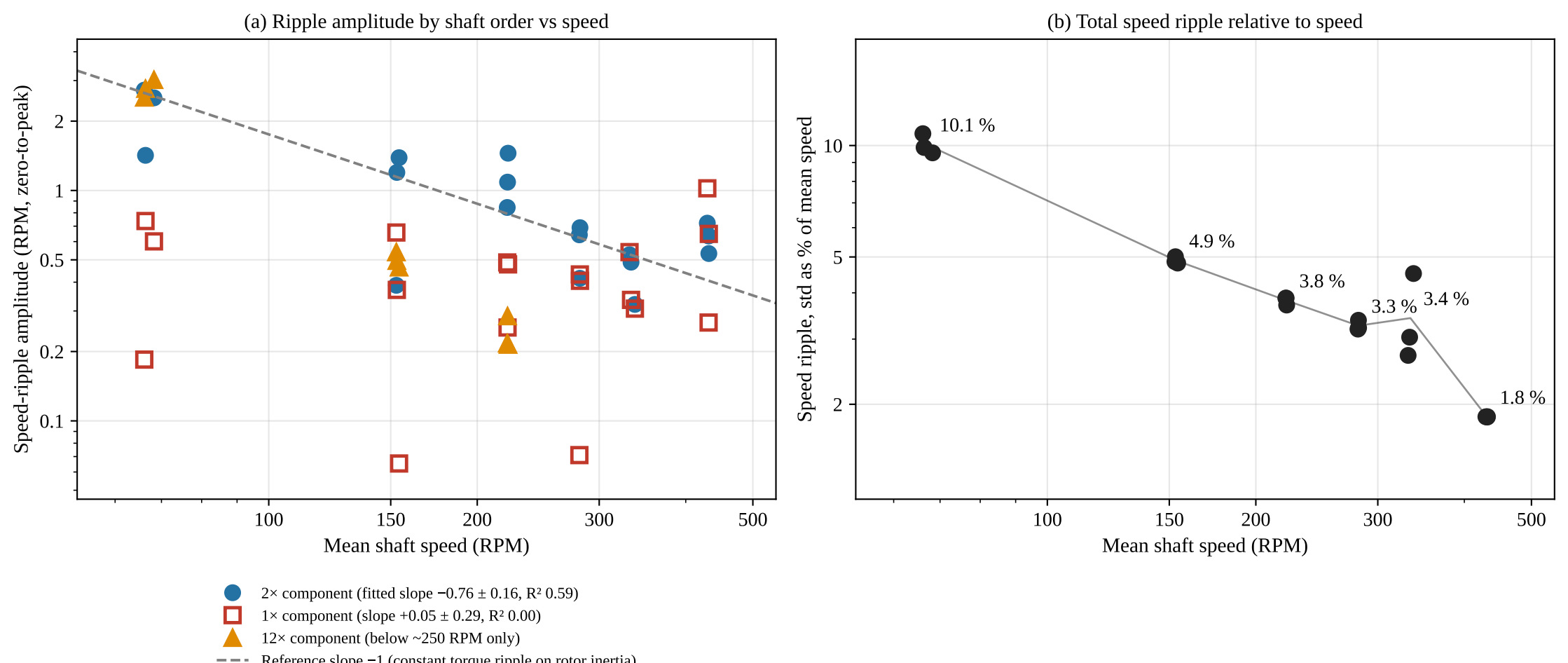
(a) Ripple amplitude by shaft order vs speed
Speed-ripple amplitude (RPM, zero-to-peak)
2
1
0.5
0.2
0.1
100
150
200
300
500
Mean shaft speed (RPM)
2× component (fitted slope −0.76 ± 0.16, R² 0.59)
1× component (slope +0.05 ± 0.29, R² 0.00)
12× component (below ~250 RPM only)
Reference slope −1 (constant torque ripple on rotor inertia)
(b) Total speed ripple relative to speed
Speed ripple, std as % of mean speed
10
5
2
10.1 %
4.9 %
3.8 %
3.3 %
3.4 %
1.8 %
100
150
200
300
500
Mean shaft speed (RPM)